

Week 2: Basic Software Development Concepts

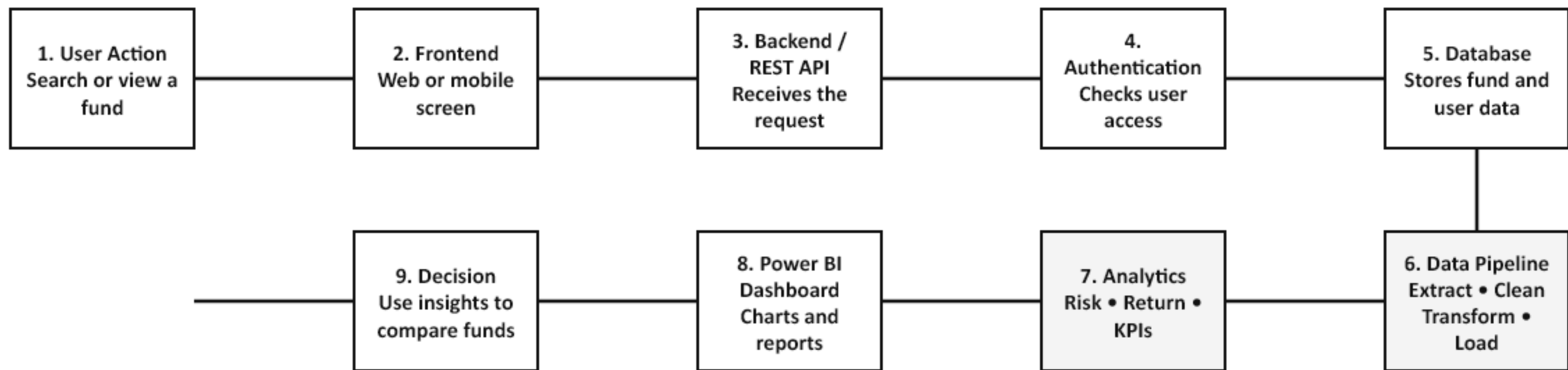
A beginner-friendly view of how a mutual-fund analytics application works

USER ACTION → DATA → INSIGHTS

Frontend • Backend • Data Pipeline • Power BI

How information moves through the application

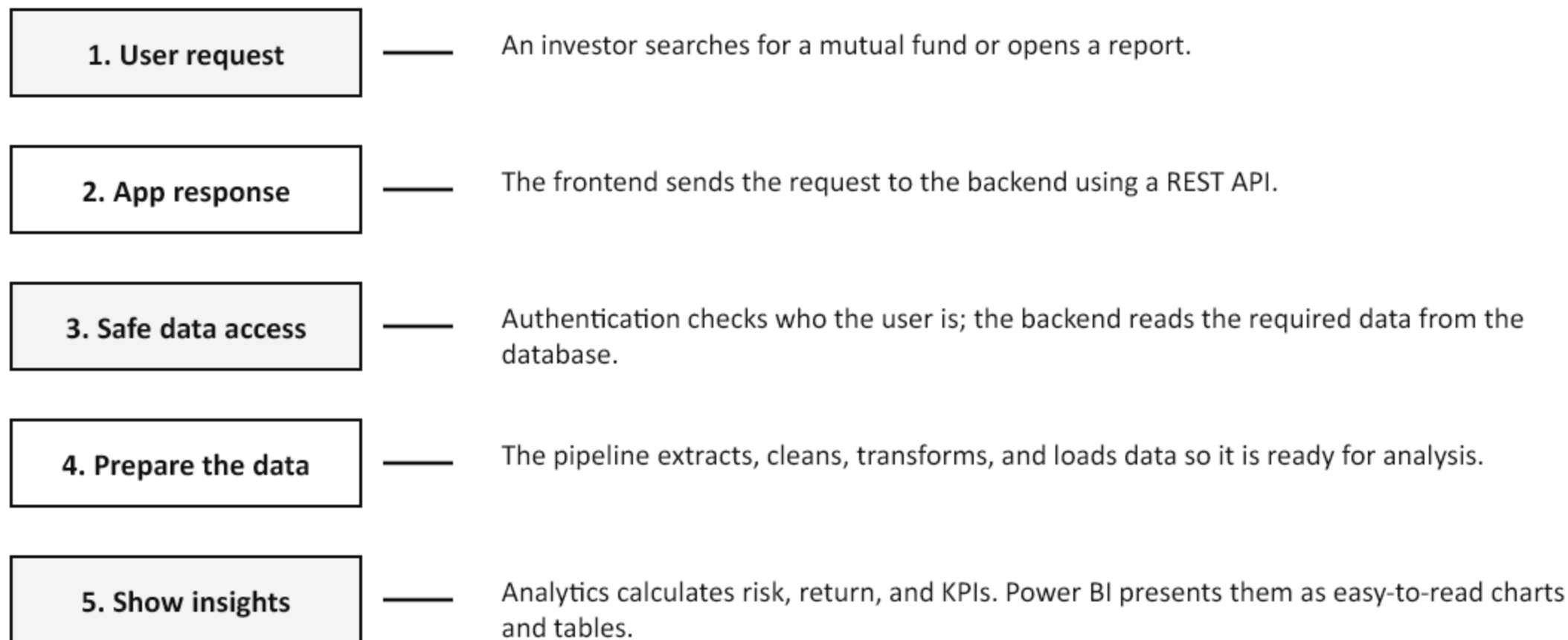
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The user starts with a simple action. The system safely processes it, turns raw fund data into useful metrics, and displays the result in a Power BI dashboard.

The data flow in simple steps

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Key idea: each layer has a specific job, making the application easier to build, test, and maintain.

Key concepts to remember

WEEK 2 | 03

Frontend	The part the user sees and clicks, such as pages, filters, and charts.
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Database	An organized place for fund details, prices, investor details, and saved reports.
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Backend	The hidden part that applies rules, handles requests, and connects to data.
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Data Pipeline	A repeatable process that moves and prepares data: Extract → Clean → Transform → Load.
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Client-Server Architecture	The client (app) asks the server for data; the server processes and returns a response.
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Logging	Recording important events, requests, and errors so issues can be investigated.
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REST API	A standard way for the frontend and backend to communicate using requests and responses.
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Error Handling	Showing a useful message and handling problems safely when something goes wrong.
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Authentication	Confirms a user identity before allowing access to private information.
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SDLC	The software development life cycle: Plan → Build → Test → Deploy → Maintain.
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A clear path from raw data to an informed investment decision

Clear roles

Each part of the application has one main responsibility.

Reliable insights

Clean data supports meaningful risk, return, and KPI calculations.

Safer access

Authentication and error handling help protect users and information.

This is the foundation for a mutual-fund analytics dashboard project.